

Venture capital investor type and the growth mode of new technology-based firms

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Abstract Independent venture capital (IVC) investors have more powerful incentives than corporate venture capital (CVC) investors to take actions that signal their capabilities (i.e. to “grandstand”). We argue that this should engender differences in the treatment effect of IVC and CVC on the mode of growth of portfolio companies. Short-term sales growth of IVC-backed firms in the period that immediately follows the VC investment should outpace that of CVC-backed firms, while we expect no difference in employment growth. We find support for these theoretical predictions on a sample of 531 Italian new technology-based firms, using several panel estimators to control for endogeneity of IVC and CVC.

Keywords Venture capital · Corporate venture capital · High-tech start-ups · Firm growth · Grandstanding

JEL Classifications D92 · G24 · L21 · L26

1 Introduction

Since the early 1990s, the economics and finance literature has been interested in assessing the impact of Venture Capital (VC) investments on firm growth (see, e.g., Jain and Kini 1995; Manigart and Van Hyfte 1999; Engel 2002; Alemany and Martí 2005; Engel and Keilbach 2007; Colombo and Grilli 2010; Puri and Zarutskie 2011). Overall, results almost unanimously show that VC-backed companies grow more than non-VC-backed companies and that this phenomenon is due to not only a *sorting effect* (i.e., the fact that VC investors select companies with the best growth prospects) but also to a positive and genuine *treatment effect* of VC investments (e.g., Baum and Silverman 2004; Bertoni et al. 2011a). However, little is known about how this treatment effect unfolds over time and which differences may be found along different measures of firm growth. In other words, a gap in the literature exists in the understanding of how, and why, VC affects the *growth mode* of portfolio companies. This is, incidentally, a common issue in the literature on the determinants of firm growth which, according to McKelvie and Wiklund (2010), has devoted too much attention to investigating *how much*, and too little on understanding *how* and *why*, firms grow.

This gap in the literature becomes particularly important once heterogeneity among VC investors is considered. It is well accepted among practitioners and academics that VC is far from being a homogeneous

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category (e.g., Wright and Robbie 1998; Cumming 2003, 2007; Sørensen 2007; Bottazzi et al. 2008). VC investors differ along several dimensions including: investment targets, screening evaluation methods, skills and competencies, governance mechanisms, and objectives. As a consequence, investments by different types of VC investors are unlikely to produce the same effects on investee firms' conduct and performance. The distinction that has attracted the most attention from scholars is the one between independent VC (IVC) and corporate VC (CVC). IVC is the dominant type of VC, especially in the US. The typical IVC firm is structured as a management company responsible for managing several pools of capital provided by institutional and individual investors. Each pool is generally organized as a legally separate limited partnership, with a management company serving as general partner and the investors serving as limited partners. Limited partners do not participate in the active management of investments (Sahlman 1990; Wright and Robbie 1998). CVC investors are instead structured as investment vehicles or business units of nonfinancial companies (e.g., Dushnitsky and Lenox 2006). The parent company provides capital and has substantial clout with respect to investors' management decisions (Gompers and Lerner 2000; Gompers 2002).

These two extremely important types of VC investors are different from several perspectives, including their objectives, investment horizons, competencies, and complementary assets for portfolio firms (e.g., Bottazzi et al. 2008). In sum, CVC investors provide to portfolio firms access to the CVC parent company's distinctive competencies and specialized assets (e.g., applied technological competencies, manufacturing resources, sales force, distribution channels) that IVC investors cannot provide (see, e.g., Maula et al. 2005; Katila et al. 2008). Hence, they may have greater beneficial effects on portfolio firms than IVC investments. However, CVC investments also pose greater appropriability hazards (Katila et al. 2008; Dushnitsky and Shaver 2009). In addition, CVC investors are likely to have strategic objectives in addition to, or even in place of, financial objectives and are likely to use the investment for "technology-window" and "competency-enhancing" purposes (see, e.g., Siegel et al. 1988; Ernst et al. 2005; Dushnitsky and Lenox 2006; Benson and Ziedonis 2009), with a scarce interest in the performance of the investee firm per se. It is difficult to predict a priori how

these differences, which have been largely emphasized by the literature, may translate into different treatment effects on the growth of investee firms. In fact, while most of these factors are likely to generate a growth differential between IVC-backed and CVC-backed firms, they do not point in the same direction. Not surprisingly, the (sparse) empirical literature on the effects of IVC and CVC investments on firm growth overall provides mixed results (see Sect. 3).

In this paper, we contribute to the literature by examining the different impact of IVC and CVC investments on firm growth from a novel perspective. We argue that differences in investment objectives due to the different *organization* of IVC and CVC investors will affect the *growth mode* of investee firms. We build this argument from the "grandstanding hypothesis" originally proposed by Gompers (1996) (see also Gompers and Lerner 2004). Contrary to CVC investors, IVC investors, especially younger and less reputable ones, have incentives to "grandstand", i.e., to take actions that signal their capabilities, so as to be able to raise new funds from investors (Gompers 1996; Lee and Wahal 2004). Because sales growth is generally considered an indicator of business success for young firms (e.g., Weinzimmer et al. 1998; Barringer et al. 2005) and is positively related to both the likelihood of an initial public offering (IPO) and the IPO valuation (Bartov et al. 2002; Bayar 2006), the incentives to grandstand will particularly be reflected in the sales *mode of growth*. Therefore, we expect IVC investments to have a larger positive short-term treatment effect on *sales growth* than CVC investments in the period that immediately follows the first round of VC financing (i.e., a short-term treatment effect), even if the long-term treatment effects of these two types of VC investment are similar. We also expect that there will be no corresponding difference across VC investment types in the time path of the treatment effect on *employment growth*.

In the empirical section of the paper, we compare the treatment effects of IVC and CVC investments on the growth in sales and employment of new technology-based firms (NTBFs), and we analyze how these effects evolve over time. Our empirical analysis is conducted on a 10-year (1994–2003) longitudinal dataset, including 531 Italian NTBFs in manufacturing and services, 47 of which are IVC- or CVC-backed firms. We estimate the treatment effects of IVC and CVC using several econometric techniques for panel

data that duly take into account the endogeneity of VC investments.

Our results show that both IVC and CVC investments have a positive and sizeable long-term treatment effect on the sales and employment growth of portfolio firms. However, the patterns of post-investment sales growth of IVC-backed and CVC-backed firms differ substantially. The effect of IVC investments on sales growth materializes immediately after the first round of VC financing, while the contribution to sales growth of CVC investments is more gradual. Moreover, the treatment effect of IVC investments on short-term employment growth is smaller than that on sales growth. In fact, we do not find any statistically significant difference between IVC-backed and CVC-backed firms in the post-investment employment growth path. These findings are consistent with the grandstanding hypothesis: IVC investors try to accelerate those dimensions of the growth of portfolio firms (notably, sales growth) that convey a signal of their own investment ability to outside potential investors.

The remainder of the paper is structured as follows. In the next section, we illustrate the theoretical background of our analysis and derive our theoretical hypotheses. In Sect. 3, we briefly illustrate the existing empirical evidence regarding the effect of IVC and CVC investments on firm growth. In Sect. 4, we describe the sample of firms and analyze the pattern of IVC and CVC investments among Italian NTBFs. In Sect. 5, we illustrate the econometric methodology and highlight the strengths and weaknesses of the different estimation techniques used in the paper. In Sect. 6, we present the results of the econometric analysis. Section 7 concludes.

2 Theory and hypotheses

As was mentioned in the Sect. 1, the VC literature has emphasized that CVC investors have *strategic objectives* in addition to, or even in place of, financial objectives. An aspect that has been quite overlooked in this literature is that IVC investors also have strategic objectives, albeit of a different nature from those of CVC investors. Contrary to IVC investors, who lack parent companies, the strategic objectives of CVC investors and in general of all captive VC investors are inherited from their respective parent companies. IVC firms normally raise capital via limited partnerships

(or the equivalent), with a lifespan generally set to 10 years (Sahlman 1990; Gompers and Lerner 1996). Typically, well before all the cash or securities in a fund are distributed to limited partners, a new fund is created. Therefore, for IVC firms, it is of paramount importance to be able to raise new capital before limited partners have realized a capital gain on their investments. The existence of information asymmetries about the investment abilities of VC firms between general partners and outside potential investors¹ as well as the high search costs for limited partners create an incentive for IVC firms to grandstand, i.e., to *take actions that signal to potential investors their ability* (Gompers 1996, p. 134), even if these actions are costly and if they do not maximize investment returns.

For CVC investors, the presence of the parent company makes grandstanding incentives far more limited. As was mentioned above, the propensity for grandstanding is essentially due to a combination of (1) the need to raise capital at fixed terms, (2) the limited access by potential and current investors to a richer information set than the mere fund's performance, and (3) the high search costs for outside investors. CVC investors use the limited partnership structure very infrequently. More important, investment managers of CVC funds have a very close relationship with their parent company, which reduces information asymmetries between the two parties and the need to signal investment ability.

We contend that differences in grandstanding incentives between IVC and CVC investors result in *different growth modes* of IVC-backed and CVC-backed firms in the immediate post-investment period. In particular, we expect IVC investments to have a positive short-term treatment effect on sales growth, which exceeds the effect of CVC investments. No such difference should be found for employment growth.

To signal their investment ability to uninformed potential outside investors, IVC firms will struggle to foster the sales growth of portfolio firms in the period that immediately follows the first VC round. First, previous studies indicate that rapid sales growth is considered an indicator of the business success of an

¹ Note that limited partners have restricted access to deal-level information because any involvement in investment management activity might jeopardize their limited liability status (see, again, Sahlman 1990; Gompers and Lerner 1996).

entrepreneurial venture, as it documents wide market acceptance of its products and services and is difficult to achieve (Weinzimmer et al. 1998; Barringer et al. 2005).² For privately owned NTBFs, rapid sales growth is then likely to positively influence investors' assessment of firm value. In the absence of a history of profits and reliable book values, investors rely on sales as an important financial driver.³ Rapid sales growth of portfolio companies also has an indirect positive effect on VC firms' reputation. IPOs are generally regarded as the most successful exit mechanism for IVC investments (Bygrave and Timmons 1992), and, more important, the number of IPO exits is often considered synonymous with VC performance (see, e.g., Jääskeläinen et al. 2006). Therefore, IVC investors, especially younger and less reputable ones, will rush investee companies toward an IPO in an effort to establish a reputation (Gompers 1996). Bayar and Chemmanur (2011) show that the probability of a NTBF undergoing an IPO and IPO market valuation depends on the assessment of outside investors that the firm has a viable business model in the product market. In turn, this assessment is positively influenced by rapid sales growth. In accordance with this view, Bayar (2006) finds that firms with higher sales growth are more likely to choose an IPO over an acquisition.

Relying on the above arguments, we expect IVC investments to have a greater short-term treatment effect on NTBFs' sales growth than CVC investments, even in the absence of any difference in the long-term treatment effect of these two types of VC investment. We hence derive hypothesis H1.

Hypothesis H1 The positive short-term treatment effect of IVC investments on firms' sales growth is greater than that of CVC investments.

² Note, however, that this view is not unanimously shared in the growth literature. For instance, Davidsson et al. (2009) show empirically that growth in sales before profits is often quite erratic, and star companies are those that are able to combine growth and profitability (i.e., they enjoy profitable growth).

³ Internet firms best epitomize this situation: "... revenues are the lifeblood of these companies and their stocks because earnings often are nonexistent. Investors want growth at Internet firms. When they can't look at earnings, they look for go-go revenue growth" (see The Wall Street Journal, Feb. 7, 2000). "This conventional wisdom is supported by empirical findings in Davis (2002), who indicates that revenues are value-relevant for Internet firms" (Bartov et al. 2002, p. 327).

One may expect that the rapid short-term sales growth of IVC-backed firms is associated with rapid short-term employment growth. In fact, these two growth indicators generally exhibit a positive correlation (e.g., Weinzimmer et al. 1998) and similar determinants, at least for NTBFs (e.g., Colombo and Grilli 2005). However, recent studies on firms' growth modes have pointed out that the concurrent validity between measures of growth of sales and employees is less than perfect (Shepherd and Wiklund 2009), and these growth indicators sometimes follow divergent paths (Delmar et al. 2003, McKelvie and Wiklund 2010).⁴ In fact, we expect the employment growth of IVC-backed NTBFs in the period that immediately follows the first VC round to fall short of their sales growth.

Hypothesis H1 states that IVC investments spur the short-term post-investment sales growth of portfolio firms more than CVC investments. Whether this accelerated sales growth is sustainable in the long term is highly uncertain. Real option theory (Dixit and Pindyck 1994) argues that, when demand is uncertain, there is an opportunity cost of making specific (i.e., irreversible) investments because of the lost option value of waiting for new information to arrive. The higher the uncertainty, the higher is this cost. When there is considerable uncertainty, firms will refrain from making specific investments to avoid the risk of incurring sunk costs if unpredicted contingencies occur (Pindyck 1988, 1993). In accordance with this argument, we expect IVC-backed NTBFs to abstain from investing in specialized assets (e.g., specialized production, logistic and distribution facilities) and human capital (e.g., dedicated sale force) in the period that immediately follows the first investment round.

Using a transaction cost lens, the literature on firms' growth modes indicates that the correlation between employment growth and sales growth is *positively* moderated by the extent of transaction costs generated by asset specificity and other sources. In other words,

⁴ For instance, Clarysse et al. (2011) provide evidence, based on case studies of high-tech start-ups, of exceptionally high sales growth combined with hardly any growth in employment in environments characterized by high stability (i.e., tight appropriability) and low complexity (i.e., easy access to complementary assets). Conversely, if the environment is stable and complex, NTBFs are inclined to enter the "market for ideas" rather than compete in the product market, and exhibit high growth in employment with almost no growth in sales.

when there are large transaction costs to using arm's-length transactions, it is more efficient for firms to integrate transactions within the boundaries of their hierarchic organization (Williamson 1975, 1985).⁵ In accordance with this argument, Chandler et al. (2009) show that employment growth closely follows sales growth when product-specific and firm-specific investments in human capital must be made for employees to gain knowledge regarding firm's products, services, processes and organizational practices; to develop personalized relationships with customers; and to provide them customized services.

Following the same logic, to the extent that firms that are experiencing rapid sales growth abstain from making specific investments in physical and human assets because of the uncertainty that surrounds future sales, it is more efficient for these firms to contract externally than to hire employees, as under these circumstances, markets outperform hierarchies.

In accordance with this argument, IVC-backed firms that are rapidly growing in sales in the immediate post-VC investment period will prefer to resort to subcontracting and outsourcing of production- and sales-related activities rather than hiring (and training) employees. It follows that the greater short-term sales growth exhibited by IVC-backed firms in comparison with their CVC-backed counterparts is unlikely to result in an increase of similar magnitude of the number of their employees. Hypotheses H2 and H3 follow.

Hypothesis H2 The positive short-term treatment effect of IVC investments on firms' sales growth is greater than that on employment growth.

Hypothesis H3 The short-term treatment effect of IVC investments on firms' employment growth is not greater than that of CVC investments.

3 The impact of IVC and CVC investments on firm growth: empirical evidence

There are very few empirical studies that have focused their analyses on the differential impact on firm

growth of IVC and CVC investments.⁶ This dearth of contributions makes it arduous to draw any definitive conclusions on the issue. For example, the cross-sectional study by Engel (2002) finds that the estimated average treatment effect of IVC investments on the growth of the employees of German firms is more than three times greater than that of CVC investments. Conversely, Audretsch and Lehmann (2004), always focusing on Germany, show that the percentage of equity capital held by both IVC and CVC investors is positively associated with the employment growth of portfolio firms in the year in which they were listed on the *Neuer Markt*. However, in this respect, their quantile regression estimates indicate that IVC investments have a greater marginal effect on low-growth firms, whereas the opposite is true for CVC investments.

The evidence regarding the time paths of the impacts of IVC and CVC investments on firm growth is even more limited. As far as we know, Chemmanur et al. (2010) is the only study that has analyzed the post-investment growth paths of IVC-backed and CVC-backed firms. More precisely, they have compared the sales growth of US IVC-backed and CVC-backed IPO firms through matching techniques. They have found that, after an IPO, CVC-backed firms enjoy greater sales growth rates than their IVC-backed counterparts. The extent of this difference decreases over time, such that, from the second year onward, the average sales growth rates of the two categories of firms are not significantly different. This evidence does not necessarily contradict our prediction of a *greater short-term treatment effect* of IVC investments on firms' sales growth in comparison with that of CVC investments. First, the results of Chemmanur et al. (2010) relate to a sample of IPO firms. IPOs are considered the most successful exit mechanism for VC investors. In light of the different objectives pursued by IVC and CVC investors, one may wonder whether these results generalize to non-IPO VC-backed firms (i.e., firms that experienced trade sales, went bankrupt or were repurchased by entrepreneurs). Moreover, to

⁵ The proposition that asset specificity is a fundamental driver of vertical integration was originally proposed by Williamson (1971).

⁶ Most empirical studies that have compared IVC and CVC investments have considered, as a measure of outcome, the likelihood of a successful exit (e.g., undergoing an IPO or being acquired), its proceeds (e.g., IPO valuation or acquisition premium), and firm performance (see, e.g., Gompers and Lerner 2000; Maula and Murray 2001; Gompers 2002; Bottazzi et al. 2008; Ivanov and Xie 2010; Chemmanur et al. 2010).

the extent that IVC and CVC investors also differ in their propensity (and ability) to bring companies to an IPO, any comparison based on a sample composed solely of IPO firms would yield biased results. Second, the use of matching techniques does not allow for reliably disentangling the sorting and treatment effects of VC investments, as it does not control for unobservables (e.g., future growth prospects) that may differently influence selection by IVC and CVC investors. More important, VC-backed firms go public a few years after the first round of VC financing (the actual span being, incidentally, endogenous). Even if the post-IPO sales growth rates of CVC-backed firms are higher than those of IVC-backed firms, the opposite may hold true in the period immediately following the first VC round. In fact, a higher growth rate of the sales of CVC-backed firms at the IPO stage is consistent with the hypothesis that, because of grandstanding, the sales of IVC-backed firms grow earlier (i.e., before the IPO) and the sales of CVC-backed firms catch up later.

4 Data and sampling process

4.1 The sample

To test our theoretical hypotheses, we use a unique longitudinal dataset for a sample of 531 Italian NTBFs observed over a 10-year period (1994–2003). Sample firms were established in 1980 or later, were owner-managed at the time of founding, and remained up to the end of 2003. Sample firms operate in the following high-tech sectors in manufacturing and services: computers, electronic components, telecommunications equipment, optical, medical and electronic instruments, biotechnology, pharmaceuticals, advanced materials, robotics, process automation equipment, software, Internet, and telecommunications services. As most sample firms were established in the second half of the 1990s, we have, on average, seven observations per firm.

The sample is composed of IVC-backed, CVC-backed and non-VC-backed firms. The way we distinguish between the two types of investors closely mirrors the prevalent definition in the literature (see, again, Sect. 1). The CVC category includes investments made by nonfinancial companies, either through affiliated financial subsidiaries and investment funds

or through informal organizational units that are generally attached to either the R&D department or the business development department. The IVC category includes investments made by independent firms that manage funds provided by outside investors. Most of the VC investors in this category are Silicon Valley-style, independent, limited-partnership firms.

Sample NTBFs participated in a survey administered in the first semester of 2004 to a population of 1,974 entrepreneurial, growth-oriented, high-tech ventures operating in Italy at the beginning of 2004 [the Research on Entrepreneurship in Advanced Technologies (RITA), 2004 directory].⁷ The construction of the RITA directory was based on several sources (see Appendix A for a detailed description of the methods by which the directory was built and the information was collected from sample firms), including those that allowed us to identify the population of Italian VC-backed NTBFs (i.e., Italian Association of Private Equity Investors, AIFI; the financial reports of IVC investors; commercial database VentureXpert). The questionnaire that was used in the survey included a specific section about VC investments, including questions on the year in which the first round of financing occurred, and the type of investors involved.⁸ As an important advantage in comparison with previous studies in the field, this survey-based procedure offered better coverage of CVC investments, especially for those deals made by small and medium-sized firms that are often neglected by secondary sources.

Our dataset has other strengths. First, to the best of our knowledge, this is the first study that uses a long

⁷ Out of the 550 NTBFs that responded to the questionnaire, we have sufficient information to estimate the econometric models illustrated in Sect. 5 for a sample of 538 firms. We further restricted the sample to 531 firms by eliminating firms invested in by captive investors other than CVC investors (e.g., bank-affiliated VC investors and government-owned VC investors) and investments syndicated by IVC and CVC investors. Both the analysis of a broader set of captive investors and the study of the effect of syndicated investments are extremely interesting research topics. However, the limited number of observations in our sample forced us to focus only on “pure” IVC and CVC investments. In this respect, note also that results obtained including syndicated deals in both the IVC- and CVC- backed groups are virtually identical to those reported in this work.

⁸ In several cases, telephone or face-to-face interviews were made with firms’ owner-managers to obtain missing information and ensure reliability of data, including those that refer to the identity of VC investors.

longitudinal dataset, including a large number of privately held NTBFs (i.e., 98% of the sample; that is, only 11 sample firms underwent an IPO), to detect the treatment effects of IVC and CVC investments on firm sales and employment growth and their time paths. Hence, our dataset is not affected by the selection bias inherent in samples exclusively composed of IPO firms. In addition, as a consequence of the process used to build the sample, the dataset includes NTBFs that are potential targets for VC investors. Lifestyle firms and other non-growth-oriented firms that would be very unlikely to be selected by VC investors are excluded from the dataset. This sampling procedure yields more precise estimates of the relevant counterfactual (i.e., the growth rate IVC-backed and CVC-backed firms would experience if they were not VC-backed) than would be possible if, for instance, we considered only VC-backed firms or included low-tech firms and high-tech lifestyle companies, which clearly have different finance needs.

Second, we implemented robust controls for the endogeneity of VC investments. On the one hand, the rather long observation period allowed for the use of

generalized method of moments (GMM) estimation techniques for panel data. On the other hand, our dataset is very informative regarding sample firms because it includes detailed firm-, industry- and location-specific information. Hence, in testing the causality relationship between VC investments and growth, we used a rich set of instruments for VC variables to control for selection based on unobservables.

Finally, in spite of the use of survey-based data, we controlled for possible survivorship bias, albeit in a partial way. In fact, as is common in survey-based studies (for an exception, see, e.g., Eckhardt et al. 2006), the sample suffers from a survivorship bias, whereby only firms that survived up to the survey date could be included in the sample. In principle, attrition might generate a sample selection bias in our estimates. In fact, we were not able to rigorously control for this selection bias; nonetheless, we were able to implement a partial control (see Sect. 6.1).

The distribution by industry of operations, geographic area of localization, and foundation date of both the 531 sample firms and the 1,974 firms included

Table 1 Distribution of sample firms and RITA directory firms by industry, geographic area and foundation date

	RITA directory		Sample	
	<i>n</i>	%	<i>n</i>	%
Industry				
Internet and telecommunications services	700	35.5	181	34.1
Software	539	27.3	159	29.9
ICT manufacturing	442	22.4	120	22.6
Biotechnology, pharmaceuticals and advanced materials	96	4.9	22	4.1
Automation and robotics	197	9.9	49	9.2
Total	1,974	100.0	531	100.0
Geographical area				
Northwest	853	43.2	250	47.1
Northeast	447	22.6	119	22.4
Centre	366	18.6	87	16.4
South	308	15.6	75	14.1
Total	1,974	100.0	531	100.0
Foundation date				
1980–1985	345	17.5	78	14.7
1986–1991	350	17.7	97	18.3
1992–1997	622	31.5	224	42.2
1998–2003	657	33.3	132	24.9
Total	1,974	100.0	531	100.0

in the RITA directory is illustrated in Table 1. χ^2 tests reveal no statistically significant differences between the distributions of the sample firms across industries and geographic areas and the corresponding distributions of the population of RITA directory firms from which the sample was drawn ($\chi^2(4) = 2.51$ and $\chi^2(3) = 3.92$, respectively). Conversely, sample firms are somewhat older than RITA directory firms, with foundation dates more (less) concentrated in the 1992–1997 (1998–2003) period ($\chi^2(3) = 32.97$). This bias does not seem to represent a significant limitation in this study. To assess the treatment effects of IVC and CVC investments on firms' growth paths, it is important to observe firms for a sufficient length of time. From this standpoint, younger firms are less

informative than older ones, as the period in which they are observed is shorter.

4.2 Patterns of IVC and CVC investments in Italian NTBFs

The distribution by industry, geographic area, foundation date and age on the first VC finance round of the IVC-backed and CVC-backed firms in our sample is illustrated in Table 2. Among the 531 firms in our sample, 23 (4.3%) obtained financing from an IVC investor and 24 (4.5%) from a CVC investor.

In our sample, the number of CVC-backed firms is slightly higher than that of IVC-backed firms. This result contrasts with the evidence provided by previ-

Table 2 Distribution of sample VC-backed firms by industry, geographic area, foundation date and age at time of first VC financing round

Industry	IVC-backed			CVC-backed		
	<i>n</i>	%	Rate	<i>n</i>	%	Rate
Industry						
Internet and telecommunications services	12	52.17	6.6	10	41.67	5.5
Software	3	13.04	1.9	6	25.00	3.8
ICT manufacturing	6	26.09	5.0	5	20.83	4.2
Biotechnology, pharmaceuticals and advanced materials	1	4.35	4.5	1	4.17	4.5
Automation and robotics	1	4.35	2.0	2	8.33	4.1
Total	23	100.00	4.3	24	100.00	4.5
Geographical area						
Northwest	11	47.83	4.4	8	33.33	3.2
Northeast	5	21.74	4.2	8	33.33	6.7
Centre	5	21.74	5.7	7	29.17	8.0
South	2	8.70	2.7	1	4.17	1.3
Total	23	100.00	4.3	24	100.00	4.5
Foundation date						
1980–1985	3	13.04	3.8	4	16.67	5.1
1986–1991	3	13.04	3.1	6	25.00	6.2
1992–1997	10	43.48	4.5	5	20.83	2.2
1998–2003	8	34.78	6.1	10	41.67	7.6
Total	23	100.00	4.3	24	100.00	4.5
Age on first VC round						
0–1	12	52.17		14	58.33	
2–5	5	21.74		4	16.67	
>5	6	26.09		6	25.00	
Total	23	100.00		24	100.00	

Rate (in percent) is the number of IVC-backed and CVC-backed firms divided by the total number of firms in the corresponding category

ous studies.⁹ There are three reasons for this discrepancy. First, in some studies, attention is exclusively focused on CVC investments made through dedicated subsidiaries, while in our dataset, these investments represent a minority of the CVC investments. Second, commercial databases used in previous studies, such as VentureXpert, rely solely on public sources of information and, hence, provide only partial coverage of CVC investments made by small and medium-sized corporations, especially when they are privately held. Finally, in contrast to the approach we followed here, previous studies consider both early stage and mature-stage investments in both high-tech and non-high-tech sectors. It has been reported that CVC investors are more oriented than IVC investors toward investments in relatively early stage high-tech firms (see, e.g., Chemmanur et al. 2010). Thus, CVC investments are relatively more common among NTBFs than among other firms. Accordingly, while focusing attention on high-tech industries, Katila et al. (2008) show that CVC investors were involved in 22% of the VC investment rounds they considered.

Let us now examine differences in the distribution of IVC-backed and CVC-backed firms. IVC and CVC investors focus on approximately the same high-tech industries ($\chi^2(4) = 4.27$), with the partial exception of software, where CVC investors are more active than IVC investors. The distributions by geographic area and date of foundation of IVC-backed and CVC-backed firms are also similar. Again, the null hypothesis that the two distributions do not differ cannot be rejected at conventional confidence levels ($\chi^2(3) = 3.71$ and $\chi^2(3) = 6.02$, respectively). In particular, both IVC and CVC investors invest little in the South of Italy, the less developed part of the country. Nearly half of both IVC-backed and CVC-backed sample firms received the first round of VC financing prior to foundation or in the first year of existence (52.2 and

58.3%, respectively). The null hypothesis that there is no difference between the two categories of VC-backed firms in firm age during the first round of VC financing cannot be rejected at conventional confidence levels ($\chi^2(2) = 0.47$).

To provide further insight into selection by IVC and CVC investors, we estimated a semi-parametric Cox (1972) competing-risks model. The dependent variables are the likelihood of the focal firm obtaining IVC or CVC financing after t years from foundation, conditional on not obtaining such financing up to time t (i.e., the hazard rate of IVC and CVC financing, assuming independent risks). The set of independent variables is the following: firm size measured as the log number of employees (full-time equivalent, including owner-managers) and lagged by one period ($LnEmployees$);¹⁰ dummies indicating whether the firm was located in a technology incubator ($DIncubated$), whether it benefited at foundation from a non-equity alliance with another firm ($DAlliance at foundation$), whether it was created by one or more academic founders ($DASU$), and whether it received a public subsidy either from the national government or from local governmental authorities in year t ($DPublic subsidy$); and the cumulative number of patents granted to the focal firm up to year t ($Stockpatent$). Patents granted are assigned to the year in which the application was made and are discounted using a 0.15 discount factor, as is customary in the literature on innovation economics (see, e.g., Griliches 1992). We also consider the human capital of the founding team, measured by founders' years of university-level economic and managerial education ($Ecoeduc$) and technical and scientific education ($Techeduc$) and by years of work experience, either in the sector of operation of the focal firm ($Spec work exp$) or in another sector ($Other work exp$). All these variables are averaged across founders. $DManager$ is a dummy variable that equals 1 if one or more of the founders previously had a managerial position in another firm. Finally, $VCArea$ and $VCSector$ capture, respectively, the relative density of VC investments in the province (i.e., NUTS3 areas) of localization and the industry of operation of the focal firm (source: the Italian association of Private Equity and Venture Capital investors, AIFI).

⁹ For instance, CVC investments represent a small share (~6%) of the 32,364 VC investments considered by Gompers and Lerner (2000). In the sample of 1,510 VC-backed IPO firms analyzed by Ivanov and Xie (2010), only 219 are backed by one or more CVC investors. Among the 11,556 VC investors considered by Chemmanur et al. (2010), there are 926 CVC investors. They were involved in only 7,180 of the 140,915 VC rounds analyzed in their study. In the sample of 750 VC investors considered by Bottazzi et al. (2008), CVC accounts for 8%; CVC investors were involved in 8.8% of the 1,643 investments under scrutiny.

¹⁰ We also estimated the model using sales as a measure of firm size. The results are very similar; $LnEmployees$ has, however, the advantage of a higher number of available observations.

The results of the estimates are listed in Table 3. There are both differences and similarities in the selection patterns of IVC and CVC investments. First, there are no significant differences in the duration dependence of the hazard rate across the two types of VC investments. The hazard rates for both IVC and CVC investments are significantly larger for younger firms and decrease rapidly with firm age (see Fig. 1).

Regarding the characteristics of the founding team, both IVC and CVC investors are more likely to invest in firms created by entrepreneurs with prior managerial experience and greater economics and management education. Moreover, the hazard rates for both IVC and CVC investments are negatively affected when firms obtain public subsidies, even though the *DPublic subsidy* variable is (weakly) significant only in the IVC equation. Quite interestingly, our estimates reveal a few remarkable differences in the patterns of IVC and CVC investments. First, IVC investors prefer to select larger firms than do CVC investors; the coefficient for firm size is of larger magnitude and is significant at 1% in the IVC estimates, whereas it is only close to statistical significance in the CVC estimates. Second, the greater the stock of patents, the higher the hazard rate of IVC investments is, which is consistent with the signaling role of quality attributed to patents (see, e.g., Baum and Silverman 2004; Hsu and Ziedonis 2007). Conversely, no CVC-backed firm in the sample was granted a patent before the first VC round (for this reason, we were forced to exclude *Stockpatent* from the CVC equation). Moreover, CVC investors are attracted by firms created by individuals with higher technical and scientific education. These findings are consistent with the view that these investors pursue “technology-window” objectives and scout for promising novel technologies at an early stage of development. Finally, CVC investments are not sensitive to the distribution of VC investments across industries and provinces, whereas *VCSector* and *VCArea* have significant positive coefficients in the IVC equation.¹¹

To summarize, our estimates indicate that IVC and CVC investors indeed invest in firms with different characteristics. However, they are consistent with the

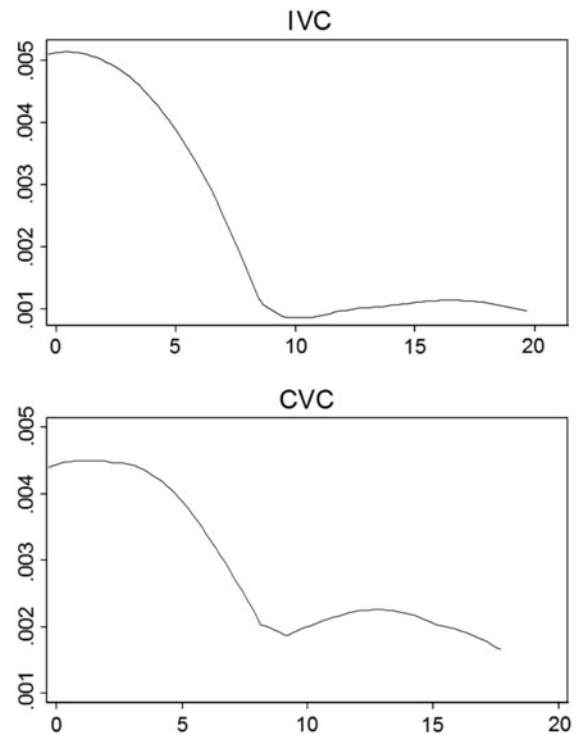


Fig. 1 Baseline hazard rate of receiving IVC and CVC investments. The figure shows the baseline hazard rate (i.e. $h_0(t)$) of obtaining IVC or CVC. The probability distribution of duration is specified by the function $F(t) = \Pr(T < t)$, which measures the probability that the duration variable T is less than some value t . The hazard function is defined as $h(t) = f(t)/S(t)$, where $f(t)$ is the probability density function and $S(t)$, which is equal to $1 - F(t)$, is the survival function. The hazard function gives the instantaneous rate of receiving, respectively, IVC or CVC financing, given that this has not happened up to t . The hazard function is modeled through a semi-parametric Cox proportional-hazard model as $h(t) = h_0(t) \exp(\beta x)$, where $h_0(t)$ is the baseline hazard rate, β a vector of parameters to be estimated, and x the set of covariates. The estimates for β are reported in Table 3

evidence provided by Gompers and Lerner (2000) and with recent evidence on patterns of VC investments in Europe showing that, at least when compared to other types of captive investors (i.e. bank-affiliated VC and governmental VC), IVC and CVC are relatively similar in their preferences (Bertoni et al. 2011b). Notably, the differences we highlight in Table 3 seem to be far less dramatic than those reported by other recent studies (see, e.g., Chemmanur et al. 2010; Ivanov and Xie 2010). A possible reason for this discrepancy is that we focus our attention on relatively young firms that operate in high-tech industries. In other words, VC investments in mature firms and in

¹¹ Apparently, these latter results diverge from those presented in Table 1. The reason is to be traced to the *ceteris paribus* nature of the multivariate econometric analysis performed herein.

Table 3 Determinants of the likelihood of receiving VC, IVC and CVC investments: Cox semi-parametric survival and competing-risks models

	VC	IVC	CVC
<i>LnEmployees(t - 1)</i>	0.7225*** (0.2093)	0.7630*** (0.237)	0.4027 (0.2607)
<i>DIncubated</i>	-0.5342 (0.5713)	-0.9768 (0.8464)	0.0584 (0.6997)
<i>DAlliance at foundation</i>	0.3445 (0.3973)	0.3371 (0.496)	0.3164 (0.6506)
<i>DASU</i>	0.348 (0.4531)	0.0025 (0.8084)	0.3395 (0.5534)
<i>Stockpatent</i>	0.2448 (0.2118)	0.3908 (0.1886)**	
<i>DPublic subsidy</i>	-1.1138** (0.5585)	-1.1773* (0.6973)	-1.5122 (1.0487)
<i>Ecoeduc</i>	0.426*** (0.0939)	0.5705*** (0.1236)	0.3124** (0.1211)
<i>Techeduc</i>	0.0963 (0.0689)	0.1124 (0.0941)	0.1739* (0.0918)
<i>Spec. work exp.</i>	0.0174 (0.0257)	0.009 (0.0365)	-0.0194 (0.0461)
<i>Other work exp</i>	0.0272 (0.0198)	0.0176 (0.0297)	0.0067 (0.0279)
<i>DManager</i>	0.9941*** (0.3104)	1.298*** (0.4267)	1.0167** (0.4603)
<i>VCarea</i>	0.2399*** (0.0853)	0.3849*** (0.1090)	-0.0446 (0.1584)
<i>VCsector</i>	0.0727*** (0.0278)	0.106*** (0.0335)	0.0343 (0.0383)
No. obs.	3,231	3,356	3,336
No. groups	512	518	513
Wald χ^2	79.60 [13]***	118.13 [13]***	19.47 [12]*
Proportional Hazard test	9.98 [13]	18.66 [13]	11.16 [12]

Robust standard errors in round brackets, degrees of freedom in square brackets. The dependent variable is the hazard rate of obtaining VC, IVC or CVC. Test of the proportional-hazard hypothesis by Schoenfeld (1982) residuals. *LnEmployees(t - 1)* is the log number of employees (full time equivalent, including owner-managers) lagged one period. In order to exploit all available information, this variable has been pre-multiplied by a dummy variable that equals 0 in the foundation year and 1 in subsequent years. *DIncubated*, *DAlliance at foundation* and *DASU* are dummy variables indicating respectively whether the firm was located in a technology incubator, whether it benefited, at foundation, from a non-equity alliance with another firm, and whether it was created by one or more academic founders. *Stockpatent* is the stock of granted patents of the firm in year *t* measured by using a 0.15 yearly discount factor (Griliches 1992). *DPublic subsidy* is a dummy variable denoting firms that in year *t* received a public subsidy either from the national government or from local governmental authorities. *Ecoeduc*, *Techeduc*, *Spec work exp*, and *Other work exp* measure respectively founders' average: years of university-level economic and managerial education; years of university-level technical and scientific education; years of previous work experience in the same sector as the firm; years of previous work experience in sectors other than that of the firm. *DManager* is a dummy variable that equals 1 if one or more of the founders previously had a managerial position in another firm. *VCArea* and *VCSector* capture the relative density of VC investments in the province (i.e. NUTS3 areas) of localization and industry of operation of the focal firm. *VCArea* is calculated as follows. Let VCA_k indicate the fraction of VC-backed firms in the k-th area of the country out of all those which were invested by VC between 1997 and 2003 (source: AIFI). Let then A_k be the fraction of Italian NTBFs in the k-th area in 2003 according to the RITA Directory. Then: $VCArea_k = VCA_k/A_k$. *VCSector* follows the same logic across sectors rather than geographic areas. CVC-backed firms did not have any patents at the time of financing, hence we were forced to exclude *Stockpatent* from the CVC model

* $p < .10$, ** $p < .05$, *** $p < .01$

firms that operate in non-high-tech industries, which are not considered in this study, are possibly responsible for most of these differences.

A last important consideration is in order. In the Italian context considered here, the incentives of IVC investors to grandstand are likely to be magnified. The VC industry has a very short history in Italy; while very few VC investments were carried out before the mid-1990s, especially in early stage firms, in the late 1990s, the number of VC investments (and investors) grew substantially (Bertoni et al. 2006). Accordingly, most IVC investors have a very short track record (if any) and need to signal their ability to uninformed outside investors to be able to raise more capital.

5 Econometric methodology

5.1 Specification of the econometric models

The impact of IVC and CVC investments on firm growth is investigated through the estimation of an augmented Gibrat-law dynamic panel data model (Chesher 1979):

$$\begin{aligned} \ln Size_{i,t} = & \alpha_1 \ln Size_{i,t-1} + \alpha_2 \ln Age_{i,t-1} + \psi_1 DIVC_{i,t} \\ & + \varphi_1 DCVC_{i,t} + W_i + \varepsilon_{i,t}. \end{aligned} \quad (1)$$

The starting point is a standard specification in the literature on industrial organization (see, e.g., Sutton 1997; Caves 1998). $\ln Age_{i,t-1}$ is the logarithm of the age of firms; $\ln Size_{i,t-1}$ is the logarithm of the size of firms. We measure size alternatively as the number of employees and sales. In this framework, the effect of size on firm growth is captured by $(\alpha_1 - 1)$. If the estimated value of α_1 is smaller than unity, firm growth decreases with increasing size. We depart from the standard specification in that we include the dummy variables $DIVC_{i,t}$ and $DCVC_{i,t}$, which indicate the VC status of firms (IVC-backed, $DIVC_{i,t} = 1$; CVC-backed, $DCVC_{i,t} = 1$). $DIVC_{i,t}$ and $DCVC_{i,t}$ switch from 0 to 1 in the year in which firms obtain the first round of IVC or CVC financing.¹² If IVC (CVC)

investments positively affect firm growth, we obtain $\psi_1 > 0$ ($\varphi_1 > 0$).¹³ Consequently, if $\psi_1 > \varphi_1$ ($\psi_1 < \varphi_1$), IVC investments have a relatively more (less) positive treatment effect on firm growth than CVC investments.¹⁴ W_i are unobservable firm-specific time-invariant characteristics, and $\varepsilon_{i,t}$ are i.i.d. error terms.¹⁵

In estimating model (1), we test whether, after receiving VC, the growth rate of the focal firm persistently increases with respect to a situation with no VC financing and whether any difference exists in the impact between IVC and CVC investments. As to this latter aspect, the main objective of our analysis is the investigation of the time path of the treatment effects of IVC and CVC investments on firms' sales and employment growth. For this purpose, we differentiate between the short-term effect of IVC and CVC investments on firm growth in the early period after the first round of VC financing (i.e., in the year in which the first VC round occurs and the subsequent year) and the long-term effect (i.e., in the steady state), and estimate the following augmented Gibrat-law dynamic panel data model:

Footnote 12 continued

additional resources. This effect is likely to persist even after the exit of the VC investor.

¹³ Note that here we investigate the impact of IVC and CVC on the growth rate of NTBFs and not their impact on the level of the dependent variables (sales and employees). If, in principle, the two issues are different and result in different conclusions (see on this point the discussion in Vanacker et al. 2011), they largely coincide in our specific case remarkably because 60% of sample firms are observed from their inception. We thank an anonymous referee for having raised the point.

¹⁴ Unfortunately, data on the amount of VC financing received by firms are not available. These data are generally regarded as confidential by owner-managers and could not be obtained from public sources (financial accounts) because the instruments used by VC investors differ (e.g., convertible bonds, straight equity). We also do not know the timing of following rounds of VC financing. We acknowledge this as a limitation of the present study.

¹⁵ Including time dummies in the models results in very erratic estimates for the parameter of $\ln Age$ (possibly because of the multi-collinearity between time dummies and $\ln Age$ in first-differenced equations). However, estimates of other parameters, most notably those corresponding to $DIVC$, $DCVC$, $DIVC2$ and $DCVC2$, are largely unaffected by the inclusion of time dummies, such that our core results are robust to this different specification.

¹² Note that these variables do not switch back to 0 when VC investors exit. The reason is that firms that obtained VC are inherently different from those that never did. For instance, VC financing is assumed to signal the quality of a firm to uninformed third parties, making it easier for the firm to obtain access to

$$\begin{aligned} LnSize_{i,t} = & \alpha_1 LnSize_{i,t-1} + \alpha_2 LnAge_{i,t-1} + \psi_1 DIVC_{i,t} \\ & + \psi_2 DIVC2_{i,t} + \varphi_1 DCVC_{i,t} \\ & + \varphi_2 DCVC2_{i,t} + W_i + \varepsilon_{i,t}. \end{aligned} \quad (2)$$

In this specification, $DIVC_{i,t}$ and $DCVC_{i,t}$ switch to unity in the year in which the first round of VC financing occurs, as in Model (1), whereas $DIVC2_{i,t}$ and $DCVC2_{i,t}$ start to equal unity in year 2, following the first round of VC financing. Therefore, whereas ψ_1 and φ_1 reflect the short-term treatment effects of IVC and CVC investments on firm growth, the positive effects in subsequent years, if any, are captured by $\psi_1 + \psi_2 > 0$ and $\varphi_1 + \varphi_2 > 0$. The distinction between short- and long-run effects is somewhat arbitrary. The 2-year lag in $DIVC_{i,t}$ and $DCVC_{i,t}$ is the shortest possible, which can provide meaningful results with respect to a short-term effect of IVC and CVC on firm growth. First, a 1-year lag would suffer from a significant measurement error. VC investments at time t eventually occur for different companies on different days within the year, such that, at time $t + 1$, the actual time since the investment ranges between little longer than 1 year to almost 2 years. Moving to a 2-year lag clearly limits the extent of this error. Second, VC investments are often followed by organizational and strategic changes, which are very unlikely to occur (and produce appreciable effects on growth) in only a few months.¹⁶

Models (1) and (2) are estimated for an unbalanced panel dataset on 531 firms observed from 1994 (or from their foundation, if after 1994) up to 2003. The dataset includes 3,686 firm-year observations on employees and 3,628 on sales.¹⁷ In Table 4, we report descriptive statistics on this sample.

5.2 Estimation methodologies

The inclusion in all models of the lagged-dependent variable as one of the covariates and the endogenous

nature of the relationship between VC investments and firm size require the use of appropriate estimation techniques. This latter aspect is crucial for the purpose of the present study. A positive association between IVC and CVC investments and firm growth may simply be the result of selection. This problem is likely to be especially significant for NTBFs because their growth performance is closely related to unobservable characteristics, such as the development of an innovative business idea or a team of smart owner-managers.¹⁸ As long as VC regressors are correlated with the error terms $\varepsilon_{i,t}$, their coefficients in a simple pooled ordinary least squares (OLS) regression are biased, and the same holds for within-group (WG or fixed-effects) estimates if the unobservable shocks are time-varying. To address endogeneity, following the literature on dynamic panel data models (Arellano and Bond 1991; Arellano and Bover 1995; Blundell and Bond 1998), we used GMM-system (GMM-SYS) estimates. Instruments in levels for endogenous variables start from $t - 2$ in differenced equations, and additional moment conditions are used with first differences as instruments for equations in levels, starting from $t - 1$ for $\Delta LSize$ and the ΔIVC and ΔCVC variables.¹⁹

This standard GMM-SYS approach enables us to control for the endogeneity of VC investments under the assumption of sequential exogeneity (see, e.g., Wooldridge 2002, p. 300). In fact, it assumes that *future* shocks to firm size are independent of VC investments, whereas *past* shocks can affect them. If VC investors can identify companies with good future

¹⁶ In this respect, note that we also tested alternative definitions (results are omitted here for brevity): the more the threshold defining the short-term effect is postponed, the more results converge to the “static” version of Eq. 1.

¹⁷ The actual number of observations included in the estimates is slightly lower due to our model specification, estimation strategy, and the presence of (few) missing data. As a matter of fact, most estimates are made on 3,045 observations for 531 firms in the employment equation and 2,998 observations for 526 firms in the sales equation.

¹⁸ If unobservables also influence the ability of firms to attract VC investors, a spurious correlation between VC investments and growth follows because of unobserved heterogeneity. An opposite bias is also possible if NTBFs with superior growth prospects self-select out of the VC market. In a thin VC market, finding a suitable offer from a VC investor might be difficult; with owner-manager time being the scarcest NTBF resource, the opportunity costs of search for a VC investor are clearly higher the better the prospects of the firm.

¹⁹ Note that we also check if the use of all available moment conditions and the consequent large number of instruments can result in significant finite-sample biases causing potential distortions as suggested by Bond (2002). To deal with this problem but also to consider possible measurement errors (see again Bond 2002), we re-estimated the model with a reduced instruments set using moment conditions in the interval between $t - 3$ ($t - 2$) and $t - 6$ ($t - 5$) for instruments in levels (differences). The results (available upon request to the authors) are very similar to those exposed in the Sect. 6.

Table 4 Descriptive statistics

Variable	Obs.	Mean	SD	Min	Max
<i>Sales</i>	3,628	2,175	8,172	0.8520	164,575
<i>Employees</i>	3,686	15.687	34.94	0.000	633
<i>Age</i>	3,686	7.348	5.724	0.000	23
<i>DIncubated</i>	3,671	0.071	0.256	0.000	1.000
<i>DAlliance at foundation</i>	3,686	0.069	0.253	0.000	1.000
<i>DASU</i>	3,686	0.098	0.297	0.000	1.000
<i>Stockpatent</i>	3,686	0.087	0.585	0.000	19.317
<i>DPublic subsidy</i>	3,686	0.204	0.403	0.000	1.000
<i>Ecoeduc</i>	3,622	0.274	0.827	0.000	5.000
<i>Techeduc</i>	3,622	1.819	2.203	0.000	8.000
<i>Spec. work exp.</i>	3,584	4.058	6.126	0.000	35.500
<i>Other work exp</i>	3,534	7.455	8.006	0.000	49.000
<i>DManager</i>	3,653	0.249	0.433	0.000	1.000
<i>VCarea</i>	3,686	1.078	1.199	0.000	5.892
<i>VCsector</i>	3,686	1.777	2.952	0.000	28.314

Sales is the level of firm sales (measured in thousand Euro). *Employees* is the number of firm employees. *Age* is firm age. *DIncubated*, *DAlliance at foundation* and *DASU* are dummy variables indicating respectively whether the firm was located in a technology incubator, whether it benefited, at foundation, from a non-equity alliance with another firm, and whether it was created by one or more academic founders. *Stockpatent* is the stock of granted patents of the firm in year t measured by using a 0.15 yearly discount factor (Griliches 1992). *DPublic subsidy* is a dummy variable denoting firms that in year t received a public subsidy either from the national government or from local governmental authorities. *Ecoeduc*, *Techeduc*, *Spec work exp.*, and *Other work exp* measure respectively founders' average: years of university-level economic and managerial education; years of university-level technical and scientific education; years of previous work experience in the same sector as the firm; years of previous work experience in sectors other than that of the firm. *DManager* is a dummy variable that equals 1 if one or more of the founders previously had a managerial position in another firm. *VCArea* and *VCSector* capture the relative density of VC investments in the province (i.e. NUTS3 areas) of localization and industry of operation of the focal firm. More specifically *VCArea* is calculated as follows. First, we considered the total number of high-tech firms that obtained VC financing over the period 1997–2003 (source: AIFI). Let VCA_k indicate the share accounted for by geographical area k out of this number. Let A_k be the estimated shares accounted for geographical area k (NUTS3 level) out of the total number of Italian NTBFs in 2003 (source: RITA Directory). Then: $VCArea_k = VCA_k/A_k$. *VCSector* follows the same logic across sectors rather than geographic areas

growth prospects that are unobservable to other investors, this assumption does not hold. To check whether the results obtained by the standard GMM-SYS model were immune to this risk, we adopted three other estimation strategies. First, we added two variables to the set of instruments of the GMM-SYS estimator that reflect the presence and activism of VC investors in the geographic area in which the focal NTBF is located ($VCArea_i$) and in the industry in which it operates ($VCSector_i$). The logic is to exploit the abundance or scarcity of VC in markets relevant to the focal firm (either the geographic or industry market) to control for the effect of selection (for a similar approach, see Bottazzi et al. 2008; Chemmanur et al. 2008; Ivanov and Xie 2010; for a detailed discussion, see Bertoni et al. 2011a). Second, as illustrated in Sect. 4.2, we estimated a selection

equation for the hazard rate of NTBFs obtaining IVC or CVC in a given year using a semi-parametric competing-risks model. We used these estimates to compute the probability of the focal NTBF receiving IVC and CVC financing in t and inserted these values as additional instruments in the GMM-SYS estimation of Eqs. 1 and 2 (for a very similar procedure, albeit in a different context, see Benfratello and Sembenelli 2006). Third, we adapted a typical Heckman (1978, 1979) two-step sample selection approach to our longitudinal setting. Estimates of the VC selection equation²⁰ were used to compute a correction term based on the Lee (1983) generalization of the Heckman (1979) two-stage selection model (see Eckhardt et al.

²⁰ The results obtained from the semi-parametric Cox survival model are exposed in the first column of Table 3.

2006 for an identical methodology for selection correction). This inverse Mills ratio type of control factor was included in a WG regression on the subset of firms that received IVC or CVC financing in the observation period (firms that did not receive VC financing were excluded from these second-stage estimates) to control for unobserved heterogeneity that affects both the selection equation and the growth equation.²¹

To evaluate the relevance of our econometric models, we applied different diagnostic tests that are standard in the GMM context. First, we implemented the Arellano and Bond test for first- and second-order serial autocorrelation of residuals [AR(1), AR(2)]. If ε_{it} is not serially correlated, the difference of residuals should be characterized by a negative first-order serial correlation and the absence of a second-order serial correlation. The Hansen test for the validity of over-identifying restrictions was implemented for each regression. This statistic tests the null hypothesis that the specified orthogonality conditions are equal to zero (Hansen 1982). Finally, we tested the strength of the additional instrument of SYS-GMM versus those of a DIF-GMM specification set by estimating the respective pseudo-first-stage regressions. The results, reported in Appendix B, strongly suggest that DIF-GMM instruments are weaker, and the SYS-GMM methodology is hence preferable.

6 Econometric results

6.1 Effect of IVC and CVC investments on firm growth

Table 5 shows the results of the estimation of Eq. 1 using the different GMM estimators described in the previous section [columns *GMM-SYS* (1)–(3)], with firm size measured as sales (Panel A) and number of employees (Panel B). In all models, the null hypothesis of the absence of a negative first-order serial correlation between differenced residuals is rejected, whereas the null hypothesis of the absence of a

second-order serial correlation is not (with one exception in the estimates of the sales equation). Hansen tests also indicate that the null hypothesis of equality to zero of the specified orthogonality conditions is not rejected. Column *HSS* reports estimates of the WG Heckman-type sample selection model.

Independently of the specific econometric technique and the measure of growth considered (sales vs. employees), the estimates of Model (1) consistently reveal that both IVC and CVC investments have a positive long-term treatment effect on the growth rate of NTBFs. The estimated coefficients for the IVC and CVC variables are always positive in both the employment and sales equations and significant at conventional confidence levels, with only one exception for CVC investment in the sales equation [*GMM-SYS* (3), see Panel A]. The coefficient for CVC is always smaller than that for IVC (except in the *HSS* estimates of the sales equation, see Panel A). However, the difference is almost never statistically significant [again, with one exception in *GMM-SYS* (3) estimates of sales equation, see Panel A].²²

Regarding control variables, the estimated coefficient for firm size is significantly smaller than unity in both the employment and sales equations. This result is consistent with the stylized fact highlighted by the empirical literature on the Gibrat law (e.g., Sutton 1997, Caves 1998) that smaller firms tend to grow faster than larger firms. The estimated coefficient for firm age generally (with one exception) is negative and not significant (again, with one exception).

As a first (important) robustness check, we tested whether these results might be somewhat driven by a survivorship bias inherent in our data. We considered an earlier release of the RITA dataset, the RITA 2000 sample. This sample, composed of 401 firms, was

²¹ The functional form of this added regressor is $\lambda_{it} = \varphi[\Phi^{-1}(F_i(t))] \cdot (1 - F_i(t))^{-1}$, where $F_i(t)$ is the cumulative hazard function for VC backing of firm i at time t , φ is the standard normal density function, and Φ^{-1} is the inverse of the standard normal distribution function.

²² The coefficient for the inverse Mills ratio control for sorting [IMR(VC)] in the WG estimates reported in column *HSS* is not significant in either the employment or sales equations. In contrast to the results for US firms, in previous studies (e.g., Sørensen 2007; Chemmanur et al. 2008), there is no evidence in our data that VC investors pick companies with the best future growth prospects. As a corollary, we are confident that selection of firms with future high-growth prospects is not the main reason for the positive relationship between IVC and CVC investments and firm growth highlighted by our estimates. For a more thorough discussion of this issue, see Colombo and Grilli (2010). For similar evidence on the absence of a positive sorting effect for European VC-backed firms, see Engel (2002) and Bottazzi et al. (2008).

Table 5 Effect of IVC and CVC investments on firm growth

	GMM-SYS (1)	GMM-SYS (2)	GMM-SYS (3)	HSS
Panel A: sales				
<i>LnSize</i> ($t - 1$)	0.7730*** (0.0602)	0.7673*** (0.0561)	0.8659*** (0.0494)	0.3212*** (0.1572)
<i>LnAge</i> ($t - 1$)	-0.1197 (0.08)	-0.1064 (0.0727)	-0.1619** (0.0696)	0.4665 (0.3064)
<i>DIVC</i> (t)	0.4866*** (0.1448)	0.4971*** (0.11439)	0.4340*** (0.1166)	0.3684** (0.1865)
<i>DCVC</i> (t)	0.2759* (0.1607)	0.2909* (0.1575)	0.1885 (0.1215)	0.5582* (0.2911)
<i>IMR</i> (VC)				-0.2468 (0.2290)
Sector dummies	Yes	Yes	Yes	No
N. obs.	2,998	2,998	2,596	209
N. groups	526	526	497	43
AR (1)	-6.17***	-6.23***	-5.93***	
AR (2)	1.53	1.53	1.93*	
Hansen test	110.99 [123]	113.42 [127]	119.80 [119]	
<i>DIVC</i> – <i>DCVC</i>	0.2106 (0.1842)	0.2061 (0.1862)	0.2455* (0.1330)	-0.1898 (0.3487)
Panel B: employees				
<i>LnSize</i> ($t - 1$)	0.8686*** (0.0495)	0.8795*** (0.0484)	0.9485** (0.0381)	0.6301*** (0.0884)
<i>LnAge</i> ($t - 1$)	-0.0147 (0.0339)	-0.0279 (0.0378)	-0.0417 (0.0476)	-0.0968 (0.1115)
<i>DIVC</i> (t)	0.3952*** (0.1264)	0.3786*** (0.1228)	0.2810*** (0.1019)	0.5318*** (0.1482)
<i>DCVC</i> (t)	0.2832*** (0.0837)	0.2682*** (0.0849)	0.2635*** (0.0866)	0.3063*** (0.0922)
<i>IMR</i> (VC)				-0.0022 (0.1752)
Sector dummies	Yes	Yes	Yes	No
<i>N</i>	3,045	3,045	2,637	214
<i>N</i> groups	531	531	502	43
AR (1)	-8.86***	-8.88**	-8.97***	
AR (2)	-0.55	-0.54	-0.30	
Hansen test	107.89 [124]	110.15 [126]	108.30 [120]	
<i>DIVC</i> – <i>DCVC</i>	0.1120 (0.1023)	0.1104 (0.1010)	0.0175 (0.1010)	0.2255 (0.1522)

Column *GMM-SYS* (1) report standard system GMM estimates. Column *GMM-SYS* (2) reports system GMM estimates with the addition, among instruments, of *VCArea* and *VCSector*. Column *GMM-SYS* (3) reports system GMM estimates with the addition, among instruments, of the estimated probabilities of a firm being IVC-backed and CVC-backed based on the competing-risks (Cox proportional-hazard) model estimates reported in Table 3. Column *HSS* reports WG (fixed-effects) regression on the subsample of VC-backed firms with the addition of *IMR*(VC), the inverse Mills ratio control for non-exogeneity in VC selection based on Cox proportional-hazard model estimates. AR(1) and AR(2) are tests of the null hypothesis of respectively no first- or second-order serial correlation. Hansen is a test of the validity of the overidentifying restrictions based on the efficient two-step GMM estimator. All GMM estimates are based on the hypothesis of IVC and CVC being endogenous, which implies their use as instruments in GMM models dated at least $t - 2$ for the equations in first differences and at least $t - 1$ for the equations in levels. *DIVC-DCVC* is the difference between the estimates of the coefficients of *DIVC* and *DCVC*. All GMM estimates are based on a two-step model with robust standard errors and finite sample correction (Windmeijer 2005). *HSS* estimates are computed with robust standard errors (White 1980). Standard errors in round brackets, degrees of freedom in square brackets. The *p*-value relating to the coefficient of *LnSize*($t - 1$) refers to the null hypothesis that its coefficient equals unity

* $p < .10$, ** $p < .05$, *** $p < .01$

selected according to the same criteria and with the same strategy used for the sample examined in the present study (Colombo et al. 2004). Thirty-one sample firms were VC-backed at the beginning of 2000 (18 IVC-backed and 17 CVC-backed; 4 firms received both types of VC investment). To assess whether a survivorship bias might influence the results of the estimates, we adapted to our specific framework a methodology that was recently proposed by Semykina and Wooldridge (2010) that extends the variable addition test for selection bias in panel data developed by Wooldridge (1995) (see also Baltagi 2003, p. 223) to a setting with unobserved heterogeneity and endogenous regressors.²³ We estimated a probit model of firm exit (due to cease of operations or acquisition) in 2000–2003 conditional on survival up to the end of 1999; the dependent variable in this model is the hazard rate of exit of sample firms in 2000–2003. The independent variables include the founders' human-capital variables, described in Sect. 4.2, receipt of VC investment before 2000, firm size and age in 1999, the level of infrastructure and resources development in the province, and industry dummies (see Table 9 in the Appendix). We used the estimated coefficients for this sample selection model to compute the inverse Mills ratio of exit for each firm–year observation included in the sample analyzed in the present work. This time-varying ratio was then inserted as a control for survivorship bias in the growth equations, which are then estimated via an instrumental variable approach, as indicated by Semykina and Wooldridge (2010).²⁴ The estimates of the standard GMM-SYS model are illustrated in Table 10 in the Appendix. The results of this test support the view that our findings are not driven by survivorship bias: the coefficient for the inverse Mills ratio control for survivorship bias is not significant in the employment and sales equations, and

the inclusion of the control for firm exit does not greatly alter the relationship between different types of VC investment and firm growth.

6.2 The time path of the effect of IVC and CVC investments on firm growth

To analyze the dynamics of the treatment effect of IVC and CVC investments on firms' employment and sales growth, we turn attention to Eq. 2. The results are reported in Table 6.²⁵

Panel A presents the results of the estimates. To gauge the magnitude of the positive effects of IVC and CVC investments on firm growth over time, Panel B of Table 6 compares the estimated effects of IVC and CVC investments T years after the first VC round with respect to the counterfactual (i.e., no reception of VC). Owing to the recursive nature of the model specification, these effects are given by a nonlinear combination of the estimated parameters. More precisely, the difference (in percentage) between the estimated size of an IVC-backed and CVC-backed firm T years after the first VC round and the estimated size a benchmark firm would exhibit in the absence of VC is equal to:

$$\hat{E}_{IVC(T)} = \psi_1 \sum_{\tau=0}^T \alpha_1^{\tau} \quad \text{if } T < 2$$

$$\hat{E}_{IVC(T)} = \psi_1 \sum_{\tau=0}^T \alpha_1^{\tau} + \psi_2 \sum_{\tau=0}^{T-2} \alpha_1^{\tau} \quad \text{if } T \geq 2. \quad (3a)$$

$$\hat{E}_{CVC(T)} = \varphi_1 \sum_{\tau=0}^T \alpha_1^{\tau} \quad \text{if } T < 2$$

$$\hat{E}_{CVC(T)} = \varphi_1 \sum_{\tau=0}^T \alpha_1^{\tau} + \varphi_2 \sum_{\tau=0}^{T-2} \alpha_1^{\tau} \quad \text{if } T \geq 2 \quad (3b)$$

Because the coefficient of the autoregressive term turns out to be statistically lower than unity ($\alpha_1 < 1$), we can also estimate the theoretical long-run effect of IVC and CVC investments on firm size using the following expression:

²³ We followed as closely as possible the above-mentioned test procedure given our data constraints. In particular, because we estimated a single probit equation for firm exit, the only major simplification with respect to Wooldridge's original framework and its subsequent extension is the assumption that the impact of each determinant of firm exit does not vary over time. The same procedure has also been employed in Colombo et al. (2009) and Bertoni et al. (2011a).

²⁴ The inverse Mills ratio in the Semykina and Wooldridge (2010) test is treated as an exogenous covariate (we thank Anastasia Semykina for personal communication confirming this as the correct procedure).

²⁵ For simplicity, we only report standard GMM-SYS estimates [corresponding to GMM-SYS (1) in Table 5]. The results of the estimates obtained using the other estimators are close to those presented here and are available from the authors upon request.

Table 6 Evolution over time of the effects of IVC and CVC investments on firm growth

	Sales	Employees				
Panel A: results of the estimates						
$LnSize(t-1)$	0.7578*** (0.0693)	0.9130** (0.0427)				
$LnAge(t-1)$	-0.1129 (0.0852)	-0.0424 (0.0501)				
$DIVC(t)$	1.0382*** (0.3415)	0.6053*** (0.1157)				
$DIVC2(t)$	-0.4755 (0.3143)	-0.3087*** (0.1005)				
$DCVC(t)$	0.3209 (0.2165)	0.4236*** (0.098)				
$DCVC2(t)$	-0.0418 (0.2977)	-0.2284** (0.1031)				
Sector dummies	Yes	Yes				
N	2,998	3,045				
N groups	526	531				
AR (1)	-6.00***	-6.09***				
AR (2)	1.35	1.52				
Hansen test	107.31 [121]	113.74 [100]				
	Sales	Employees				
	IVC CVC IVC-CVC	IVC CVC IVC-CVC				
Panel B: effects of IVC and CVC investments on firm growth over time						
$T = 0$	1.0382*** (0.3415)	0.3209 (0.2165)	0.7172* (0.3723)	0.6053*** (0.1157)	0.4236*** (0.098)	01817 (0.1379)
$T = 1$	1.8249*** (0.5720)	0.5641 (0.3728)	1.2608* (0.6467)	1.1579*** (0.2118)	0.8103*** (0.1798)	0.3476 (0.2638)
$T = 2$	1.9456*** (0.5464)	0.7066* (0.3634)	1.2390** (0.6092)	1.3537*** (0.2554)	0.9349*** (0.1865)	0.4188 (0.2873)
$T = 3$	2.037*** (0.5758)	0.8146* (0.4407)	1.2225* (0.6815)	1.5325*** (0.3077)	1.049*** (0.2050)	0.4879 (0.3370)
$T = 4$	2.1064*** (0.6226)	0.8964* (0.5278)	1.2100 (0.7804)	1.6958*** (0.3687)	1.1527*** (0.2284)	0.5431 (0.3975)
$T \rightarrow \infty$	2.3233*** (0.8565)	1.1524 (0.8553)	1.1709 (1.2126)	3.4084** (1.3512)	2.2429*** (0.6860)	1.1655 (1.3379)

Estimates reported in Panel A refer to Eq. 2. All estimates are based on a two-step system GMM model with robust standard errors and finite sample correction (Windmeijer 2005). AR(1) and AR(2) are tests of the null hypothesis of respectively no first- or second-order serial correlation. Hansen is a test of the validity of the overidentifying restrictions based on the efficient two-step GMM estimator. Estimates are based on the hypothesis of $DIVC$ and $DCVC$ being endogenous, which implies their use as instruments in GMM models dated at least $t-2$ for the equations in first differences and at least $t-1$ for the equations in levels. The p value relating to the coefficient of $LnSize(t-1)$ refers to the null hypothesis that its coefficient equals unity. Panel B reports the estimated effect of IVC and CVC, as described in equations (3) and (4), based on estimates reported in Panel A by using the Delta method for standard errors. $IVC-CVC$ is the difference between the estimated effect of the coefficients of IVC and CVC. Standard errors in round brackets, degrees of freedom in square brackets

* $p < .10$, ** $p < .05$, *** $p < .01$

$$\hat{E}_{IVC(\infty)} = \lim_{t \rightarrow \infty} \hat{E}_{IVC(T)} = \frac{\psi_1 + \psi_2}{1 - \alpha_1} \quad (4a)$$

$$\hat{E}_{CVC(\infty)} = \lim_{t \rightarrow \infty} \hat{E}_{CVC(T)} = \frac{\varphi_1 + \varphi_2}{1 - \alpha_1}. \quad (4b)$$

Remember that parameters ψ_1 and φ_1 determine the effects of IVC and CVC investments in the year in which VC financing is first obtained and in the following year (i.e., the short-run effects) with respect to the counterfactual (i.e., no reception of VC). The effects from year 2 onward depend on the sum of parameters ψ_1 and ψ_2 , and φ_1 and φ_2 , for IVC and CVC investments, respectively. The statistical significance of the effects computed from expressions (3) and (4) (and their difference) can be tested using χ^2 tests performed by the Delta method (for a similar approach in a different context, see Maliranta 2005; Bertoni et al. 2010).

Our estimates highlight significant differences in the growth modes of IVC- and CVC-backed companies. In the year after VC is first obtained, the estimated sales of an IVC-backed firm are 182% greater than in the absence of IVC investment; this effect is significant at 1%. Consistently with hypothesis H1, the estimated positive effect on sales growth of CVC investments in the corresponding period is much smaller. As a consequence, there is a large wedge between the effect on firm sales of IVC and CVC investments (+126% in year 1, significant at

conventional confidence levels). Nonetheless, from the second year following receipt of the first VC round, this wedge stabilizes and is equal to +117% in the steady state, and from the fourth year onward, it is no longer significant. These results highlight the different growth mode of the sales of IVC-backed and CVC-backed firms in the post-investment period. After the initial boost, sales of IVC-backed firms remain almost constant beginning in the second year following receipt of the first VC round: the estimated steady-state treatment effect is equal to +232%. Conversely, sales of CVC-backed firms continue to grow gradually, and, beginning in year 2, the positive treatment effect of CVC investments on sales growth is (weakly) significant.

The results clearly indicate that both IVC and CVC investments boost employment growth immediately following receipt of the first round of VC financing. The estimated increases in the number of employees attributable to IVC and CVC investments at the end of the year following that of the first VC round are +116 and +81%, respectively, with respect to the counterfactual (i.e., no reception of VC). From the second year onward, the magnitude of this effect continues to increase but at a decreasing rate. In the steady state, it is equal to +341 and +224%, respectively. Consistent with hypothesis H2, the estimated effect of IVC investments on firms' employment growth immediately following the first VC round is smaller than the

Table 7 Evolution over time of the ratio between value added and sales

	All sample		VC-backed only	
	OLS	WG (fixed-effects)	OLS	WG (fixed-effects)
<i>DIVC(t)</i>	−0.1057*** (0.0401)	−0.1979*** (0.0567)	−0.1225** (0.0479)	−0.1979*** (0.0576)
<i>DCVC(t)</i>	0.0873*** (0.0329)	0.1175 (0.0779)	0.1007** (0.0438)	0.1175 (0.0791)
Constant	0.3370*** (0.0104)	0.2906*** (0.0049)	0.2962*** (0.0397)	0.2893*** (0.0327)
Sector dummies	Yes	No	Yes	No
Firm fixed-effects	No	Yes	No	Yes
<i>N</i>	1,960	1,960	206	206
<i>N</i> groups	288	288	34	34

The dependent variable is the ratio between value-added and sales (winsorized at the 1% threshold). The first two columns are estimated on the whole sample while the third and fourth column only refer to the sample of IVC and CVC-backed firms. Robust standard errors are used for both OLS and fixed-effects estimates. Standard errors in round brackets

* $p < .10$, ** $p < .05$, *** $p < .01$

corresponding effect on sales growth. Moreover, the estimated difference between the treatment effects of IVC and CVC investments on employment growth, although positive, is never significant at conventional confidence levels, as predicted by hypothesis H3.

As a further test of the extent to which our results are driven by the incentives of IVC investors to grandstand, we investigated the effects of IVC and CVC investments on the degree of vertical integration of portfolio firms. In accordance with our hypotheses, we expect IVC-backed firms to increase outsourcing rather than recruit employees after receiving VC to sustain sales growth. Outsourcing should translate into a reduction of the ratio of value added to sales. No such reduction should instead be observed in CVC-backed firms. To test this prediction, we computed the ratio of value added to sales for sample companies²⁶ and estimated OLS and WG (fixed-effects) models to check the extent to which the ratio changes after the investment. We run the estimates on the full sample (which due to data limitations is limited to 288 companies, 34 of which are VC-backed) and on the sample restricted to VC-backed firms only (34 firms). The results are reported in Table 7. Consistent with our expectations and regardless of the sample and estimation methodology, we observed a significant and sizeable reduction in the ratio of value added to sales after an IVC investment, while no significant negative effect was found after a CVC investment.

In summary, our results are consistent with our predictions. IVC investors, having incentives to grandstand, spur the sales growth of their portfolio firms in the period immediately following the first VC round. However, this sales growth does not lead to employment growth of a comparable magnitude. CVC investors do not need to grandstand. Accordingly, the short-term treatment effect of CVC investments on sales growth is smaller than that of IVC investments and is similar in magnitude to the treatment effect on employment growth. Moreover, the difference in the treatment effect on sales growth of IVC and CVC investments tends to vanish in the long term.

7 Discussion and conclusions

The aim of this paper was to analyze the impact of IVC and CVC investments on the growth mode of portfolio companies. For this purpose, we considered a longitudinal dataset for a large sample of Italian NTBFs, most of which are privately held. The long longitudinal dimension of the dataset and the availability of a rich set of primary information on sample firms allowed us to estimate an augmented Gibrat-law-type dynamic panel data model using several estimation techniques that duly take into account the endogenous nature of IVC and CVC investments.

Our results highlighted some similarities and some interesting differences between IVC- and CVC-backed firms. Both IVC and CVC investments have a positive treatment effect of similar magnitude on the employment growth of portfolio firms in the period immediately following the first VC round. From year 2 following the first VC round, the estimated number of employees for both types of firms continues to increase compared with growth in the absence of VC, but at a decreasing (and again similar) rate. The dynamic pattern and magnitude of this effect are similar to those highlighted by Puri and Zarutskie (2011) for US privately held VC-backed firms. Regarding sales growth, the positive short-term treatment effect of IVC investments is much larger than that on employment growth. In the second year following the first investment round, sales growth of IVC-backed firms decelerates. Conversely, the positive effect of CVC investment on the sales growth of portfolio firms is more gradual. In particular, in the year in which VC is first obtained and in the following year, the magnitude of this effect is significantly smaller than that of IVCs and is comparable with that on employment growth. By year 4 following the first VC round, the gap in sales growth between IVC- and CVC-backed firms, although positive, is no longer significant.

We interpret these results to be a consequence of the incentives of IVC investors to *grandstand* to be able to raise more capital. Rapid sales growth is an indicator that a firm has a viable, successful business; it makes it easier for a firm to undergo an IPO and is likely to lead to boost IPO proceeds. Therefore, IVC investors have incentives to spur the growth of the sales of their portfolio firms in the period immediately following the first investment. As it is highly uncertain

²⁶ Because the ratio is extremely leptokurtic, we winsorized the variable at the 1% threshold. Similar results are obtained using 2 and 5% thresholds.

whether this accelerated growth pace is sustainable in the long term, IVC-backed firms refrain from hiring a large number of employees to support sales growth and prefer to resort to outsourcing of production, sales and distribution. Conversely, because of their captive organization, CVC investors do not have similar incentives to grandstand. Accordingly, the growth paths of sales and employees of CVC-backed firms in the post-investment period proceed more gradually and follow similar tracks.

This study provides an original contribution to the stream of the VC literature that has considered the different value added to portfolio firms exerted by different types of VC investors. Previous studies in this branch of research that have compared the effect on firm growth of IVC and CVC investments have found mixed results (Engel 2002; Audretsch and Lehmann 2004; Chemmanur et al. 2010), which is consistent with the view that IVC and CVC investors have relative strengths and weaknesses. In this work, we offer new insights into the role of different types of VC investors in fostering the growth of NTBFs. We claim that the difference in *organization* of these two types of VC investors engenders different incentives to fuel the growth of sales, but not of employment of portfolio firms. As far as we know, this is the first study to make this link.

This work is also an original addition to the entrepreneurship literature on the growth of young ventures. McKelvie and Wiklund (2010) contended that, despite self-evident interest and massive empirical work, theoretical development in this field has been surprisingly slow. In their view, scholars have devoted too much attention to the question of *how much* firms grow, while they have failed to provide adequate answers to the question of *how* and *why* firms grow. Moreover, there are important qualitative differences between different growth indicators, which need to be carefully considered (see also Shepherd and Wiklund 2009). In this study, we made a step forward in this direction. We showed that the growth path of sales of IVC-backed firms in the post-investment period diverges from that of employment, while these two growth indicators follow a similar track for CVC-backed firms. We also provided a sound theoretical explanation for this discrepancy.

We are aware that this study has some limitations, which open new promising directions for future research. First, the question arises regarding whether

our results relating to the different impacts of IVC and CVC investments on firms' sales and employment growth paths can be generalized to countries other than Italy. In most countries outside the US, the UK and Israel, the VC industry is quite underdeveloped. To the extent that IVC investors lack a solid reputation and have incentives to grandstand, we expect the treatment effect of IVC investments on firm growth to be similar to the one highlighted in this study. Replication of this study in countries that differ as to the development of the VC industry would clearly be of great interest. Second, previous studies have documented that there is substantial heterogeneity within the two categories of investors considered here, according to characteristics such as investment experience and human capital of the managing partners (see, e.g., Sørensen 2007; Bottazzi et al. 2008). To the extent that these characteristics influence the perceptions that uninformed outside investors have of the investment management ability of VC investors, they will also (negatively) influence the incentives of IVC investors to grandstand. It would then be interesting to assess whether these characteristics differently moderate the treatment effects of IVC and CVC investments on the growth path of firms' sales. Following a similar line of reasoning, one may also expect that the reputations developed over time by different types of VC investors differently affect their treatment effects on firm growth. Another issue which would deserve specific attention, but that the limited size of our sample prevents us from studying, is the extent to which different goals of IVC and CVC investors may engender conflicts between these two types of investors. We find that patterns of investments of IVC and CVC in young high-tech companies are similar, albeit CVC seems to prefer younger and smaller firms. There is some evidence that this result is not a specificity of our sample but may be generalized at least to Europe as a whole (Bertoni et al. 2011b). Having the same "taste" for companies, IVC and CVC investors could then naturally end up co-investing. However, to the extent to which IVC and CVC may have different goals, frictions could arise about how to steer a firm's strategic decisions. The likelihood of a conflict, as well as its gravity, will depend not only upon firm and investor characteristics but also on corporate governance and contractual arrangements.

In spite of these limitations, we believe this study advances our understanding of the heterogeneity

across VC investors and the influence that different types of organization of independent and captive VC investors exert on the development of portfolio firms.

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Appendix A: The RITA directory and the sample

The sample we use in this work is representative of a population of 1,974 entrepreneurial growth-oriented high-tech ventures operating in Italy at the beginning of 2004 (2004 RITA directory). With data on 1,974 high-tech start-ups, this directory is the most detailed and comprehensive source of data on Italian NTBFs. The use of official statistics in this domain is not possible for several reasons. First, in Italy, most individuals who are defined as self-employed by official statistics (i.e., “independent employees”) are actually salaried workers with atypical employment contracts. Unfortunately, on the basis of official data, such individuals cannot be distinguished from owner-managers of a new firm. This means that the official number of NTBFs is enormously inflated, especially in sectors like software where atypical employment contracts are very popular. In addition, official data do not distinguish firms that were established by one or more entrepreneurs (i.e., owner-managed firms) from firms that were created as subsidiaries of other firms. This again inflates the number of NTBFs. Lastly, there are no official statistics about M&As: therefore one cannot distinguish firms that were acquired by another firm and lost independence while keeping their legal status, from independent NTBFs.

In absence of reliable official statistics, the RITA directory developed at Politecnico di Milano is presently the most complete source of information on Italian NTBFs. The directory was created in 2000 and it was updated in 2002 and 2004. For its

construction, several sources were used. These included: (1) the lists of the companies that are members of the national entrepreneurial associations of the focal industries; (2) the lists of the members of the regional sections of the Italian entrepreneurial association (Confindustria); (3) the lists of the members of the local Chambers of Commerce; (4) the lists of companies that participated in the most important industry trades and expositions; and (5) the lists of companies that purchased advertising services in popular off-line (e.g., *Kompass*) and on-line (e.g., *Infoimprese.it*) directories. Moreover, the RITA directory includes: (6) the population of young firms that were granted by the Italian communication authority (AGCOM) a license to provide telecommunications services (including Internet access services), (7) the population of NTBFs that were incubated in a science park or a business innovation center (BIC) affiliated with the respective national associations, (8) the population of NTBFs that obtained equity financing from VC investors that adhere to the Italian financial investor association (AIFI), and (9) the population of VC-backed NTBFs that were included in *VentureXpert*. Lastly, information provided by the national financial press, specialised magazines, and other sectoral studies was also used in the compilation of the directory. Altogether, the 2004 release of the RITA directory comprises 1,974 firms that complied with the criteria relating to industry of operations, age and independence mentioned in Sect. 3.1. For each firm, the name of a contact person (i.e., one of the owner-managers) is also provided. While the RITA directory obviously is not exhaustive of any self-employment episode in high-tech sectors, it provides an extensive and accurate coverage of the population of Italian entrepreneurial ventures in this domain excluding lifestyle companies, non-growth oriented firms and other scarcely entrepreneurial acts. In particular, it is quite unlikely that potential candidates for VC investment are excluded from the RITA directory.

In the first semester of 2004, a questionnaire was sent to the contact person of the RITA directory firms either by fax or by e-mail. The first section of the questionnaire poses detailed questions relating to the human capital characteristics of the firms' founders. The second section comprises further questions concerning the characteristics of the firms including access to VC financing, the identity of VC investors, receipt of public subsidies, and the evolution over time

of the firms' employees. Answers to the questionnaire were checked for internal coherence by educated personnel and were compared with information obtained from public sources (i.e. the firms' websites and annual reports). In several cases, telephone or face-to-face follow-up interviews were made with the

Hypothesis that all β coefficients in difference and all β coefficients in levels are jointly zero. The results are reported in the following table where we also report the adjusted R^2 of the pseudo-first-stage regression.

See Table 8.

Table 8 Pseudo-first-stage regressions

	IVC		CVC	
	Difference	Levels	Difference	Levels
$H_0 : \forall \tau : \beta_\tau = 0$	0.07	1,286***	4.45***	987.6***
Adj. R^2	0.001	0.810	0.023	0.780

The first line reports F statistics on the null Hypothesis of weak instruments

* $p < .10$, ** $p < .05$, *** $p < .01$

firms' owner-managers. This final step was crucial in order to obtain missing data and ensure that data were reliable. In addition, financial and economic data including the evolution over time of the firms' sales from 1994 onwards, and data on patent activity during the firms' entire life were obtained from public sources (i.e. the AIDA and CERVED databases and the databases of patent offices, respectively). Data on VC financing was cross-checked with those available from public and commercial sources.

Appendix B: Pseudo first-stage regressions

The orthogonality conditions of the GMM-SYS estimator imply that other than solely using lagged levels as instruments for difference equations (as in the GMM-DIF approach), lagged differences are used as instruments for level equations. The validity of these latter instruments has been verified by means of the Hansen test. However, for these instruments to be good, we have to verify that they are also not weak. In order to do so, we run the following pseudo-first-stage regressions:

$$\begin{aligned} \text{Difference: } \Delta DIVC_{i,t} &= \alpha \text{LnSize}_{i,t-2} + \sum_{\tau=t-2}^{t-6} \beta_\tau \Delta DIVC_{i,\tau} \\ &+ \gamma \Delta \ln(\text{age})_{i,t} + \Delta \omega_{i,t} \\ \text{Levels: } DIVC_{i,t} &= a \Delta \text{LnSize}_{i,t-1} + \sum_{\tau=t-1}^{t-5} \beta_\tau \Delta DIVC_{i,\tau} \\ &+ \gamma \ln(\text{age})_{i,t} + \eta_{i,t} \end{aligned}$$

Similar equations describe the pseudo-first-stage equations for $DCVC$. We then perform test the null

Lagged instruments in first differences are strongly correlated with the VC variables, whereas oppositely, lagged instruments in levels are poorly correlated with the change in the VC-backing status, pointing to the strength other than the validity of the additional instruments used in the GMM-SYS estimates.

Appendix C: Control for survivorship bias

See Tables 9 and 10.

Table 9 Probit model on exit between 2000 and 2003

	Coefficients
<i>N. founders</i>	−0.007 (0.044)
<i>Ecoeduc</i>	0.055 (0.071)
<i>Techeduc</i>	0.013 (0.040)
<i>Spec. work exp.</i>	0.024* (0.013)
<i>Other work exp.</i>	0.021** (0.011)
<i>DManager</i>	0.068 (0.178)
<i>Employees in 1999</i>	0.001 (0.002)
<i>VC</i>	0.084 (0.236)
<i>Infrastructure</i>	0.007** (0.003)

Table 9 continued

	Coefficients
<i>Age in 1999</i>	−0.011 (0.018)
<i>Constant</i>	−2.924*** (0.705)
<i>Sector dummies</i>	Yes
<i>N</i>	391

The dependent variable is a dummy equal to 1 if the firm exits from the sample between 2000 and 2003. *N. founders* measures the number of founders of the focal firm. *Ecoeduc*, *Techeduc*, *Spec work exp*, and *Other work exp* measure respectively founders' average: years of university-level economic and managerial education; years of university-level technical and scientific education; years of previous work experience in the same sector as the firm; years of previous work experience in sectors other than that of the firm. *DManager* is a dummy variable that equals 1 if one or more of the founders previously had a managerial position in another firm. *Employees in 1999* is the number of employees of the firm at the end of 1999. *VC* is a dummy variable equal to 1 if the focal firm received VC financing before 2000. *Infrastructure* is a composite index reflecting the level of infrastructure and resources development in the province of firm's location (source: Centro Studi Confindustria; see also Colombo et al. 2011). *Age in 1999* is firm's age at the end of 1999. Robust standard errors in round brackets

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 10 continued

	Sales	Employees
<i>DIVC – DCVC</i>	0.2896 (0.1885)	0.0576 (0.1164)

IMR(Survived) is an inverse Mills ratio control for survivorship bias and is obtained by the estimation of a probit model on firm exit estimated on firm exited in the 2000–2003 period conditional on survival up to the end of 1999 (see Table 9). This time varying ratio is inserted as a control for survivorship bias in the growth equations which are estimated via SYS-GMM (Semykina and Wooldridge 2010). Both estimates are based on a two-step system GMM model with robust standard errors and finite sample correction (Windmeijer 2005) under the hypothesis of *DIVC* and *DCVC* being endogenous, which implies their use as instruments since $t - 2$ for the equations in first differences and $t - 1$ for the equations in levels. *AR(1)* and *AR(2)* are tests of the null hypothesis of respectively no first- or second-order serial correlation. *Hansen* is a test of the validity of the overidentifying restrictions based on the efficient two-step GMM estimator. The p-value relating to the coefficient of *LnSize(t - 1)* refers to the null hypothesis that its coefficient equals unity. *DIVC-DCVC* is the difference between the estimates of the coefficients of *DIVC* and *DCVC*. Standard errors in round brackets, degrees of freedom in square brackets

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Table 10 Robustness check on survivorship bias

	Sales	Employees
<i>LnSize (t - 1)</i>	0.7389*** (0.074)	0.8789** (0.0539)
<i>LnAge (t - 1)</i>	−0.0649 (0.1375)	−0.0534 (0.0592)
<i>DIVC(t)</i>	0.4777*** (0.1318)	0.4739*** (0.1657)
<i>DCVC(t)</i>	0.1881 (0.1966)	0.4164*** (0.1081)
<i>IMR(Survived)</i>	0.1387 (0.9342)	−0.7349 (0.5983)
<i>Sector dummies</i>	Yes	Yes
<i>N</i>	2,880	2,927
<i>N groups</i>	507	512
<i>AR (1)</i>	−5.91***	−8.52***
<i>AR (2)</i>	1.61	−0.55
<i>Hansen test</i>	103.61 [115]	101.34 [116]

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